

ESBJERG SAEDENSTRAND, Denmark

WMO: 060890

Lat: 55.500N

Lon: 8.400E

Elev: 11

StdP: 101.19

Time Zone: 1.00 (EUC)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.4	-7.3	-12.3	1.3	-9.9	-9.4	1.7	-6.6	22.7	5.7	21.2	6.2	4.3	20	0.838

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.3	25.3	19.6	23.0	18.2	20.9	17.2	20.2	24.4	19.0	21.8	17.8	20.0	5.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.5	21.7	17.7	12.7	20.7	16.7	11.9	19.4	58.0	24.4	53.8	21.7	50.2	20.1	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 19.1	15.3	13.8		-10.2	28.5	4.7	2.4	-13.6	30.2	-16.3	31.6	-19.0	33.0	-22.4	34.7	
(5)			DB	-10.4	21.9	4.6	1.2	-13.8	22.7	-16.5	23.4	-19.1	24.1	-22.5	25.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	1.4	1.0	3.3	6.4	11.2	13.9	16.5	16.3	13.5	10.4	5.7	3.4
	DBStd	6.24	4.71	3.75	2.90	2.64	3.04	2.78	2.47	2.19	1.92	2.39	3.10	2.67
	HDD10.0	1219	266	252	209	113	19	1	0	0	1	23	131	204
	HDD18.3	3565	524	486	467	358	221	137	70	70	146	245	379	463
	CDD10.0	721	0	0	0	6	57	117	201	197	106	37	1	0
	CDD18.3	27	0	0	0	0	1	4	13	8	1	0	0	0
	CDH23.3	176	0	0	0	0	7	29	87	47	5	0	0	0
	CDH26.7	32	0	0	0	0	0	6	16	11	1	0	0	0
(14) Wind	WSAvg	7.1	8.0	7.2	7.3	6.4	6.4	6.6	7.3	7.2	7.5	7.5	6.8	7.3
(15) Precipitation	PrecAvg	782	62	41	49	38	44	53	63	71	86	97	98	79
	PrecMax	1108	144	125	118	79	192	135	174	170	204	240	174	149
	PrecMin	520	4	1	5	1	10	4	12	3	13	22	44	10
	PrecStd	137	28	30	29	21	34	30	35	38	47	52	32	39
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	8.1	11.6	17.8	24.5	27.2	29.0	28.2	22.3	17.1	12.3	9.6
		MCWB	7.2	7.6	9.0	11.8	17.7	19.0	21.4	20.7	18.6	15.3	11.3	9.0
	2%	DB	7.2	7.3	9.1	15.2	21.5	23.1	25.7	24.1	18.6	15.4	10.7	7.9
		MCWB	6.6	6.6	7.8	10.9	15.2	17.6	19.9	19.7	15.6	14.1	10.0	7.2
	5%	DB	6.7	6.6	7.7	12.2	19.1	20.9	23.4	21.2	17.2	14.2	10.0	7.2
		MCWB	6.2	5.9	6.9	9.6	14.0	16.7	18.7	18.3	15.1	13.0	9.4	6.6
	10%	DB	6.3	5.8	6.8	10.3	16.6	18.6	20.8	19.6	16.3	13.3	9.4	6.7
		MCWB	5.8	5.2	6.2	8.3	12.9	15.4	17.6	17.3	14.6	12.4	8.8	6.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	7.7	9.8	12.8	18.4	20.4	22.2	21.5	18.6	15.8	11.6	9.1
		MCDB	7.6	7.9	11.5	16.4	23.4	25.6	27.0	26.0	21.2	16.7	11.9	9.6
	2%	WB	6.8	6.8	8.0	11.3	16.2	18.4	20.5	20.0	16.6	14.5	10.2	7.6
		MCDB	7.0	7.1	8.8	14.4	19.9	21.6	25.0	23.8	17.6	15.0	10.5	7.8
	5%	WB	6.4	6.1	7.0	9.9	14.7	17.0	19.1	18.8	15.7	13.4	9.6	6.8
		MCDB	6.6	6.5	7.5	12.1	17.9	20.1	22.2	20.9	16.6	13.8	9.9	7.0
	10%	WB	5.9	5.2	6.3	8.6	13.2	15.8	17.8	17.7	14.9	12.7	8.9	6.3
		MCDB	6.2	5.6	6.8	10.1	16.0	18.2	20.2	19.2	15.9	13.1	9.2	6.6
(35) Mean Daily Temperature Range	5% DB	MDBR	3.0	3.2	3.5	5.2	6.1	5.5	5.3	4.8	4.3	3.8	3.5	2.9
		MCDBR	2.3	2.6	5.1	9.0	11.2	10.3	11.2	8.8	6.0	4.9	3.4	2.8
	5% WB	MCWBR	2.4	2.5	4.0	5.7	6.3	5.9	6.0	5.4	3.9	3.6	3.0	2.8
		MCWBR	2.3	2.2	4.3	8.3	10.1	9.0	9.8	8.0	4.9	4.2	3.2	2.6
(40) Clear-Sky Solar Irradiance	taub		0.325	0.342	0.364	0.397	0.388	0.385	0.398	0.400	0.380	0.359	0.336	0.312
		taud	2.248	2.300	2.301	2.259	2.350	2.380	2.359	2.361	2.399	2.380	2.301	2.246
	Ebn at Noon		575	720	797	821	854	860	839	812	776	694	560	499
		Edh at Noon	58	78	98	118	114	112	112	106	89	72	55	46
(44) All-Sky Solar Radiation	RadAvg		0.49	1.11	2.43	4.07	5.32	5.61	5.25	4.21	2.89	1.49	0.61	0.36
	RadStd		0.08	0.15	0.23	0.47	0.52	0.43	0.54	0.30	0.29	0.18	0.09	0.04

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)		+0.37	N/A	N/A	N/A	N/A	N/A	-117	-161	N/A	N/A

Nomenclature: See separate page